EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 1

PRINTABLE PRACTICE WORKSHEET

EDUSAMBAM, working on building your Foundational Concepts. Visit edusambam.com for Auto-graded Self-assessment Experience.

YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP

PHYSICAL

1. Download any practice test →
2. Solve within assigned time →
3. Submit it to your Teacher/Examiner →
4. Get assessed and improve

ONLINE

1. Visit edusambam.com →
2. Open Assessment Arena →
3. Solve online papers →
4. Get auto-assessed and improve

Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
School / Institution:		**Test Centre:**	

Subject	Biology	**Topic**	Biology Comprehensive Practice Test 1	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. Which statement best explains why biology is considered an integrative science? **Basic · 2 pts**

A. It studies only the structure and anatomy of the human body
B. It focuses mainly on naming and classifying organisms into groups
C. It connects processes across molecular, cellular, organismal and ecological levels
D. It avoids using any ideas from chemistry, physics or mathematics

Answer: ☐A ☐B ☐C ☐D

2. Which structure is the basic unit of life? **Basic · 2 pts**

A. Cell
B. Tissue
C. Organ
D. Ecosystem

Answer: ☐A ☐B ☐C ☐D

3. A cell membrane is especially important because it: **Basic · 2 pts**

A. Stores all of the cell's hereditary information in its own DNA
B. Builds every protein that the cell needs by itself
C. Completely blocks all substances from entering or leaving the cell
D. Controls the movement of substances between the cell and its surroundings

Answer: ☐A ☐B ☐C ☐D

4. Which sequence moves from smaller to larger levels of biological organisation? **Basic · 2 pts**

A. Organ → tissue → cell → organism
B. Cell → tissue → organ → organism
C. Organism → organ → tissue → cell
D. Ecosystem → organism → organ → cell

Answer: ☐A ☐B ☐C ☐D

5. An ecosystem includes: Basic · 2 pts

A. Only the animals and birds that live together in a habitat
B. Only the plants, fungi and microorganisms found in a habitat
C. A single population made up of one species only
D. Living organisms together with their physical environment and their interactions

Answer: ☐A ☐B ☐C ☐D

6. Which example is a population? Intermediate · 2 pts

A. All the frogs of one species living in a pond
B. All the organisms living in a pond
C. A frog and the water around it
D. All the ecosystems on Earth

Answer: ☐A ☐B ☐C ☐D

7. Biodiversity can include variation: Intermediate · 2 pts

A. Only between different ecosystems
B. Within genes, among species and among ecosystems
C. Only among individual humans
D. Only in body size

Answer: ☐A ☐B ☐C ☐D

8. A hypothesis is strongest scientifically when it: Intermediate · 2 pts

A. Cannot be tested by any experiment
B. Is based only on personal opinion
C. Produces predictions that can be tested
D. Is accepted by everyone without evidence

Answer: ☐A ☐B ☐C ☐D

9. Homeostasis refers to: Intermediate · 2 pts

A. Permanent and unlimited growth of an entire organism
B. Regular movement of organisms between different ecosystems
C. Evolution of a species over time
D. Regulation of internal conditions within suitable ranges

Answer: ☐A ☐B ☐C ☐D

10. Natural selection can change a population when: Intermediate · 2 pts

A. Individuals consciously choose which useful mutations to develop
B. Heritable variation affects survival and reproductive success in a particular environment
C. All individuals in the population are genetically identical
D. Traits are never passed on from parents to their offspring

Answer: ☐A ☐B ☐C ☐D

11. Which observation is quantitative? Advanced · 2 pts

A. The leaf looks darker
B. The fish behaved unusually
C. The plant grew 4.2 cm in seven days
D. The solution became cloudy

Answer: ☐A ☐B ☐C ☐D

12. A food web is more informative than a simple food chain because it: Advanced · 2 pts

A. Shows interconnected feeding relationships among organisms
B. Shows only a single feeding relationship in a habitat
C. Leaves out the decomposers and the producers
D. Measures exact population sizes

Answer: ☐A ☐B ☐C ☐D

13. If a habitat loses a major producer, the first major ecological effect is likely to involve: Advanced · 2 pts

A. Immediate creation of a new species
B. No change, because consumers make their own food
C. An increase in the numbers of every population
D. Reduced energy entering the food web

Answer: ☐A ☐B ☐C ☐D

14. Which statement best connects evolution and biodiversity? Advanced · 2 pts

A. Evolution removes all variation among living organisms over time
B. Evolution contributes to diversification and differences among lineages
C. Biodiversity prevents evolution from ever occurring in a habitat
D. They are unrelated processes

Answer: ☐A ☐B ☐C ☐D

15. A strong biological explanation of a declining frog population should first: Advanced · 2 pts

A. Gather evidence and consider interacting factors such as habitat, disease, predation and climate
B. Assume a single cause and look only for evidence that supports that idea
C. Ignore environmental conditions and focus on the animals alone
D. Use a single population size without any time frame or comparison

Answer: ☐A ☐B ☐C ☐D

Section B — True / False — 20 marks

16. All living organisms are multicellular. Basic · 2 pts

A. True
B. False

Answer: ☐A ☐B

17. Cells maintain organised internal processes using energy and information. Basic · 2 pts

A. True
B. False

Answer: ☐A ☐B

18. A community includes organisms of different species living and interacting in an area. **Basic · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

19. An ecosystem contains only living components. **Basic · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

20. Homeostasis means that every internal condition remains perfectly constant. **Basic · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

21. A scientific conclusion can be revised when strong new evidence appears. **Intermediate · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

22. Natural selection acts on populations across generations, not because individuals deliberately choose to evolve. **Intermediate · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

23. Biodiversity refers only to the number of species in a habitat. **Advanced · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

24. Energy generally enters many ecosystems through producers and is transferred through feeding relationships. **Advanced · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

25. A correlation by itself always proves that one biological variable causes another. **Advanced · 2 pts**

A. True **B.** False

Answer: ☐ **A** ☐ **B**

Section C — Fill in the Blank **20 marks**

26. The basic structural and functional unit of life is the _____. **Intermediate · 2 pts**

Answer: _____

27. The maintenance of relatively stable internal conditions is called _____. **Intermediate · 2 pts**

Answer: _____

28. A group of organisms of the same species living in an area is a _____. **Intermediate · 2 pts**

Answer: _____

29. The variety of life at genetic, species and ecosystem levels is called _____. **Intermediate · 2 pts**

Answer: _____

30. Organisms that make organic food from inorganic sources are called _____. **Intermediate · 2 pts**

Answer: _____

31. A proposed explanation that can be tested is a _____. **Advanced · 2 pts**

Answer: _____

32. The process by which populations change across generations is called _____. **Advanced · 2 pts**

Answer: _____

33. The biological study of interactions between organisms and their environment is called _____. **Advanced · 2 pts**

Answer: _____

34. A group of different tissues working together to perform a specific function forms an _____. **Intermediate · 2 pts**

Answer: _____

35. A network of interconnected food chains is a _____. **Advanced · 2 pts**

Answer: _____

Section D — Column Matching — 10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Cell	____	**A.** Regulation of internal conditions
Population	____	**B.** Basic unit of life
Ecosystem	____	**C.** Variety of life at multiple levels
Homeostasis	____	**D.** Organisms plus their physical environment
Biodiversity	____	**E.** Members of one species in an area

Section E — Short Answer — 20 marks

41. Explain two reasons why a cell must regulate what enters and leaves it. **Advanced · 4 pts**

42. Distinguish a population, community and ecosystem using one consistent example. **Advanced · 4 pts**

43. Explain how natural selection can increase the frequency of an inherited trait without an organism intentionally changing its genes. **Advanced · 4 pts**

44. A pond loses much of its plant life. Predict two likely consequences and explain the links. **Advanced · 4 pts**

45. Why is evidence from more than one biological level often useful when explaining a complex problem? **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

FOR OFFICE USE ONLY	
Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 1 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. C
2. A
3. D
4. B
5. D
6. A
7. B
8. C
9. D
10. B
11. C
12. A
13. D
14. B
15. A

Section B — True / False

16. False
17. True
18. True
19. False
20. False
21. True
22. True
23. False
24. True
25. False

Section C — Fill in the Blank (accepted answers)

26. cell / cells
27. homeostasis
28. population / populations
29. biodiversity / biological diversity
30. producers / producer / autotrophs / autotroph
31. hypothesis
32. evolution
33. ecology
34. organ / organs
35. food web / food webs

Section D — Column Matching

36. Cell — B
37. Population — E
38. Ecosystem — D
39. Homeostasis — A
40. Biodiversity — C

Section E — Short Answer (Model Answers)

41. A cell must regulate what crosses its membrane to keep its internal conditions stable and suitable for chemical reactions, and to take in needed substances such as nutrients and oxygen while removing wastes and keeping harmful substances out.
Marking: 2 marks: keeps internal conditions stable (homeostasis); 2 marks: lets in needed substances, removes wastes or keeps harmful substances out.

42. Using a pond as the example: a population is all the frogs of one species in the pond; a community is all the populations of different species living there (frogs, fish, plants, microbes); an ecosystem is that community together with the non-living environment (water, light, temperature) and their interactions.
Marking: 1 mark: population = one species in an area; 1 mark: community = populations of different species living together; 1 mark: ecosystem = community plus the non-living environment; 1 mark: uses one example consistently (e.g. a pond or forest).

43. Individuals in a population vary in inherited traits. If a trait helps them survive and reproduce in their environment, they leave more offspring that inherit it, so over generations the trait becomes more common. Individuals do not choose or change their genes on purpose.
Marking: 1 mark: individuals in a population vary; 1 mark: the trait is inherited; 1 mark: individuals with the helpful trait survive and reproduce more; 1 mark: the trait becomes more common over generations.

44. With fewer plants, less food and energy enter the food web, so herbivores decline, and predators that eat them then decline too. Less shelter and less oxygen released by plants may also reduce animal populations, while dead plant material may temporarily increase decomposers.
Marking: 2 marks: less food/energy for herbivores (primary consumers), so they decline; 2 marks: a second effect, such as fewer predators, less oxygen, or loss of shelter.

45. Molecular, cellular, organism and ecosystem levels are connected, so a problem seen at one level can have causes at another. Evidence from several levels gives a fuller and more reliable explanation than one level alone. For example, gene variation, cell function and habitat change together can explain why a population declines.
Marking: 1 mark: names more than one level (genes, cells, organisms, ecosystems); 1 mark: one level alone gives an incomplete explanation; 1 mark: the levels are connected and affect each other; 1 mark: combining evidence gives a stronger explanation, or gives an example.